

American Samoa Model Checks

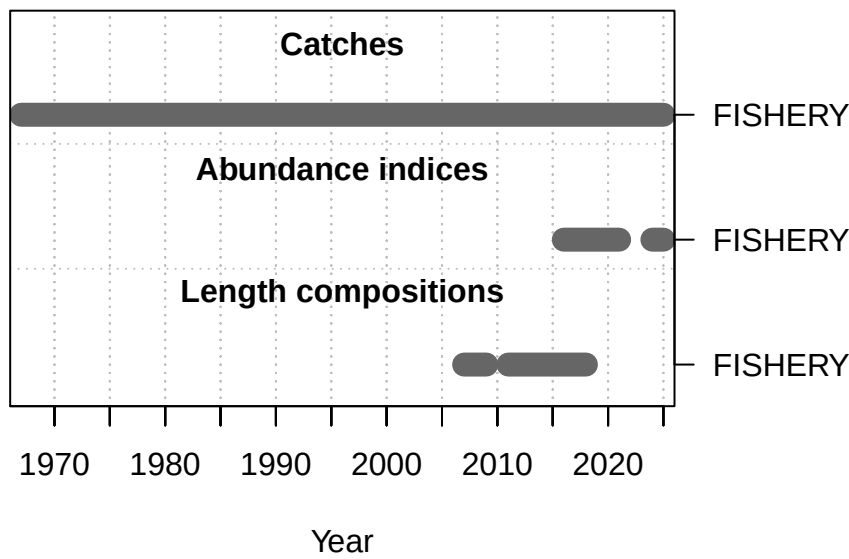
Marc Nadon and Meg Oshima

2026-04-06

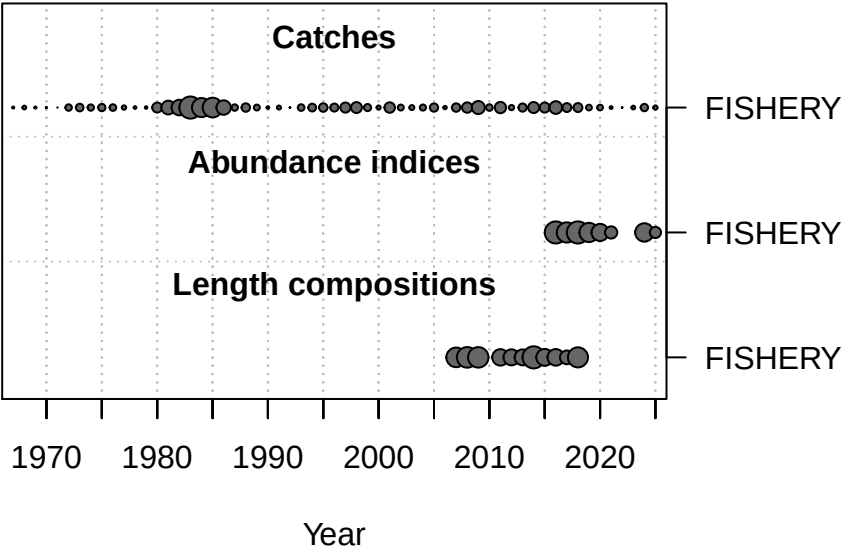
This is a summary report for the ETCO base model run.

Model Output

Input Data



Fleet	RMSE.perc	Nobs
FISHERY	131.8	8
Combined	131.8	8



Convergence Check

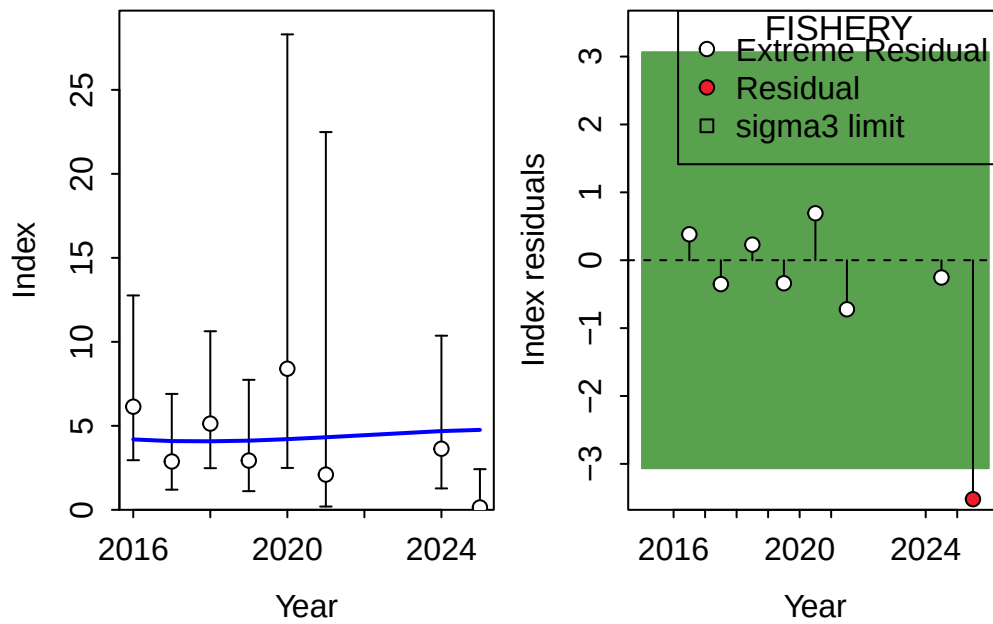
```
Converged      MaxGrad
1      TRUE 7.28816e-06
```

```
[1] "1 NOTE:  Max data length bin: 90  < max pop len bins: 100; so will accumulate larger pop
[2] "N warnings: 1"
```

Fleet	RMSE.perc	Nobs
FISHERY	6.5	11
Combined	6.5	11

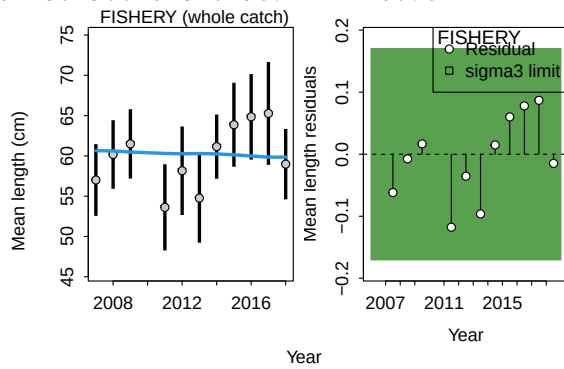
Fit to Model

CPUE

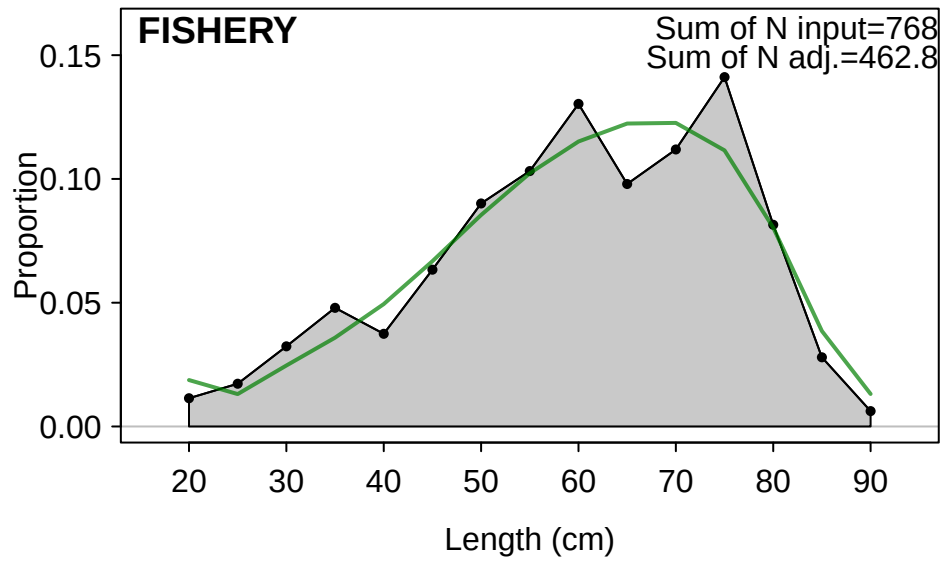


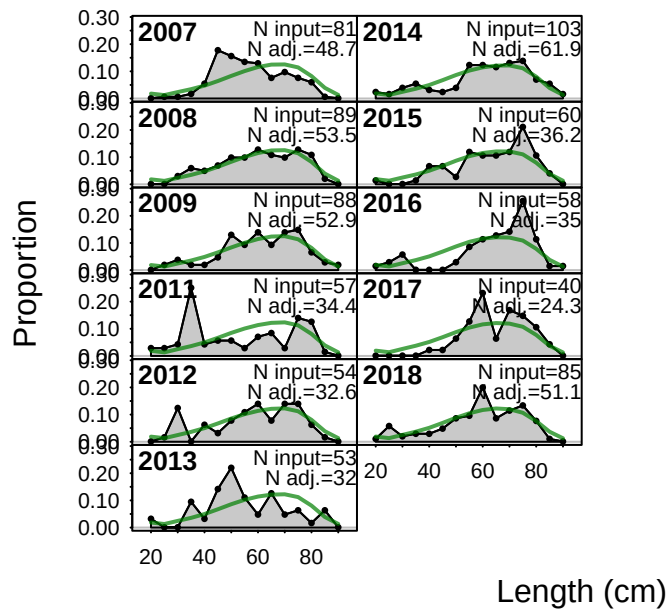
w lo hi
0.4501966 0.3164907 1.1120670

Index runs.p test sigma3.lo sigma3.hi type
1 FISHERY 0.175 Passed -0.1703368 0.1703368 len



Length Comp

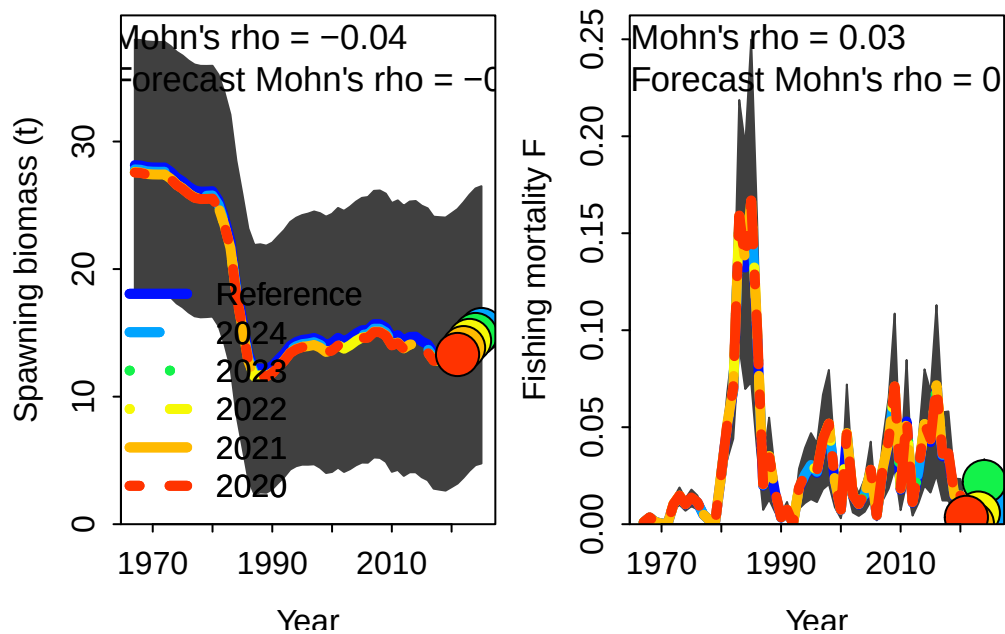




Retrospective and Hindcasting

Retrospective

Mohn's Rho stats, including one step ahead forecasts:



Mohn's Rho stats, including one step ahead forecasts:

	type	peel	Rho	ForecastRho
1	F	2024	0.02016332	0.01992353
2	F	2023	0.03394396	0.03351140
3	F	2022	0.03485873	0.03394396
4	F	2021	0.03589909	0.03485873
5	F	2020	0.03926740	0.03833676
6	F Combined		0.03282650	0.03211488

Hindcasting

Plotting Hindcast Cross-Validation (one-step-ahead)

Computing MASE with only 3 of 5 prediction residuals for Index FISHERY

Warning: Unequal spacing of naive predictions residuals may influence the interpretation of

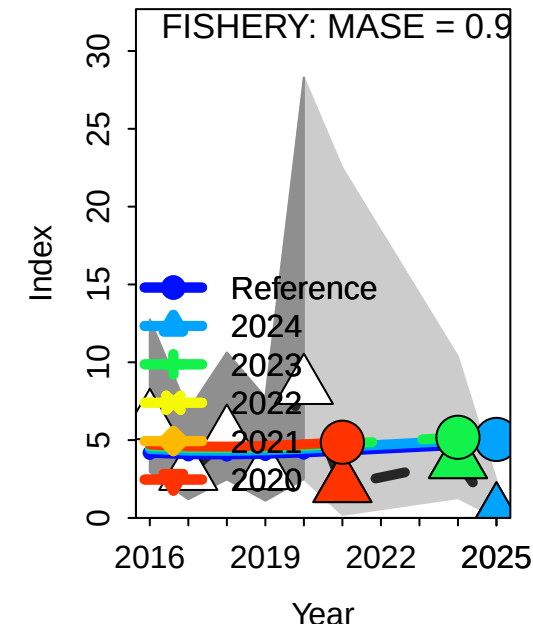
MASE stats by Index:

Plotting Hindcast Cross-Validation (one-step-ahead)

No observations in evaluation years to compute prediction residuals for Index FISHERY

MASE stats by Index:

	Index	Season	MASE	MAE.PR	MAE.base	MASE.adj	n.eval
1	FISHERY	1	NA	NA	NA	NA	0



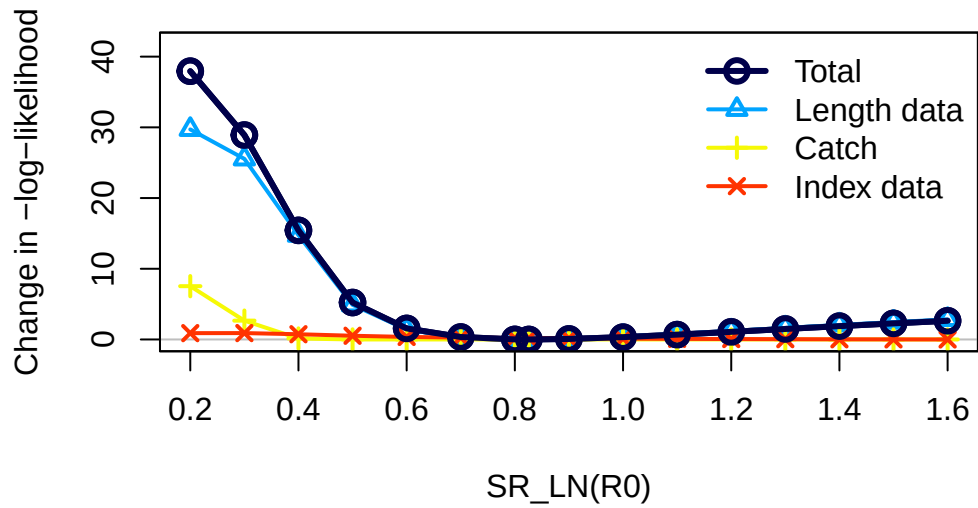
Recruitment Deviations

Likelihood Profile

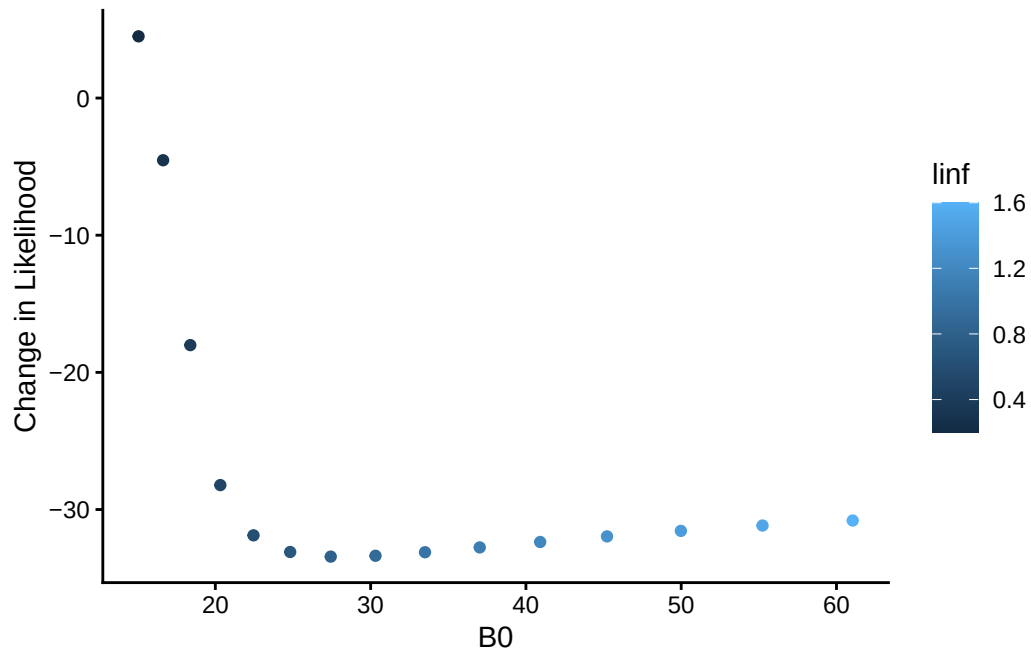
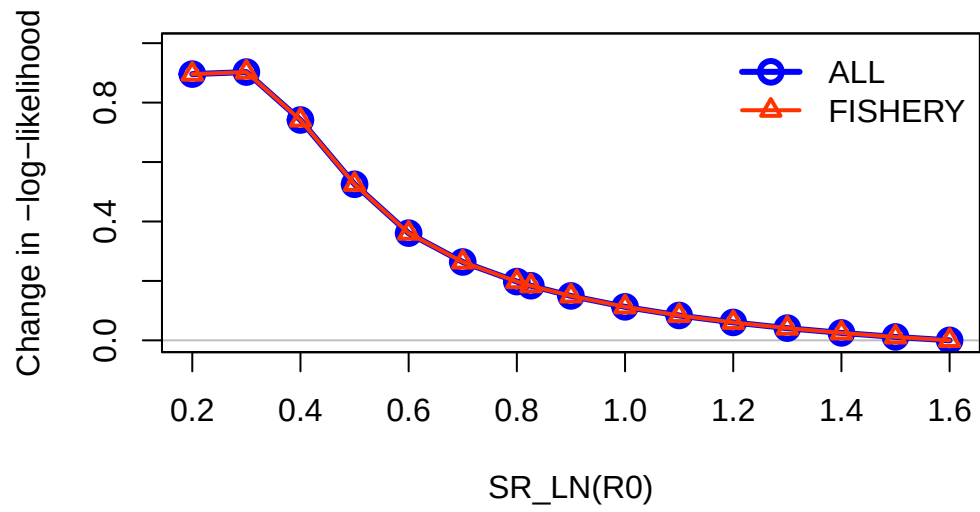
[1] "SR_LN(R0)"

	frac_change	include	label
TOTAL	1.0000	TRUE	Total
Catch	0.1986	TRUE	Catch
Equil_catch	0.0000	FALSE	Equilibrium catch
Survey	0.0238	TRUE	Index data
Length_comp	0.7830	TRUE	Length data
Recruitment	0.0000	FALSE	Recruitment

InitEQ_Regime	0.0000	FALSE	Initital equilibrium recruitment
Forecast_Recruitment	0.0000	FALSE	Forecast recruitment
Parm_priors	0.0006	FALSE	Priors
Parm_softbounds	0.0003	FALSE	Soft bounds
Parm_devs	0.0000	FALSE	Parameter deviations
Crash_Pen	0.0000	FALSE	Crash penalty

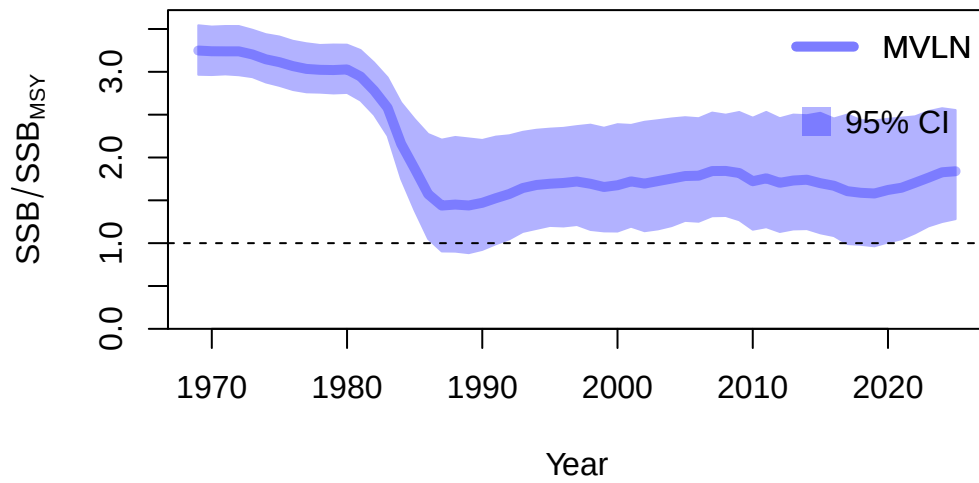
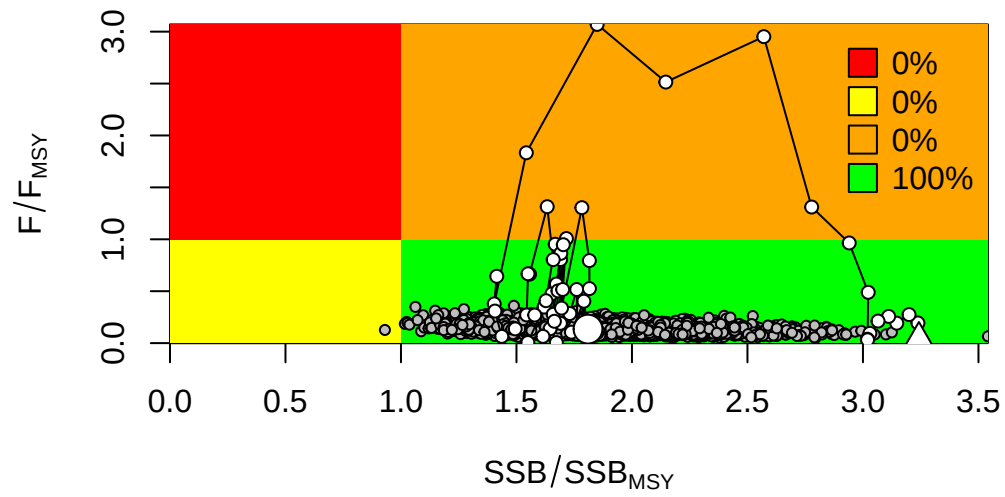


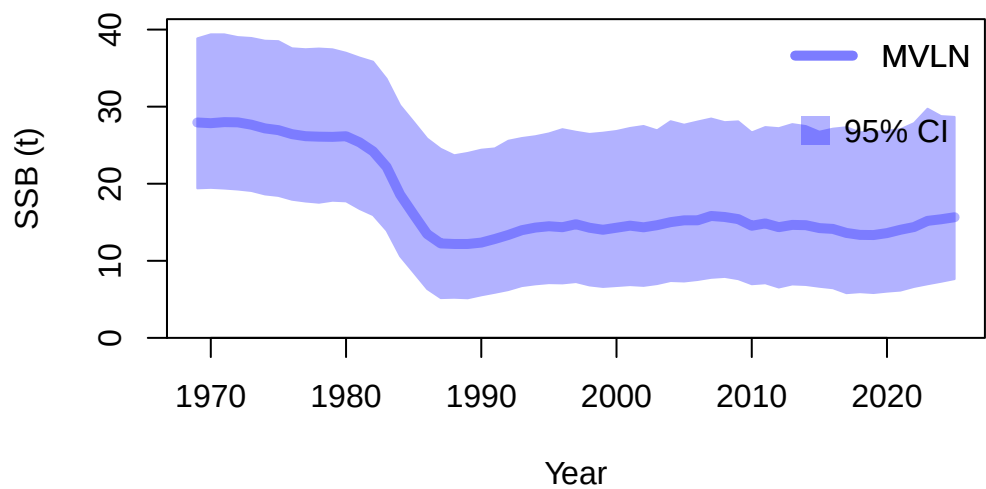
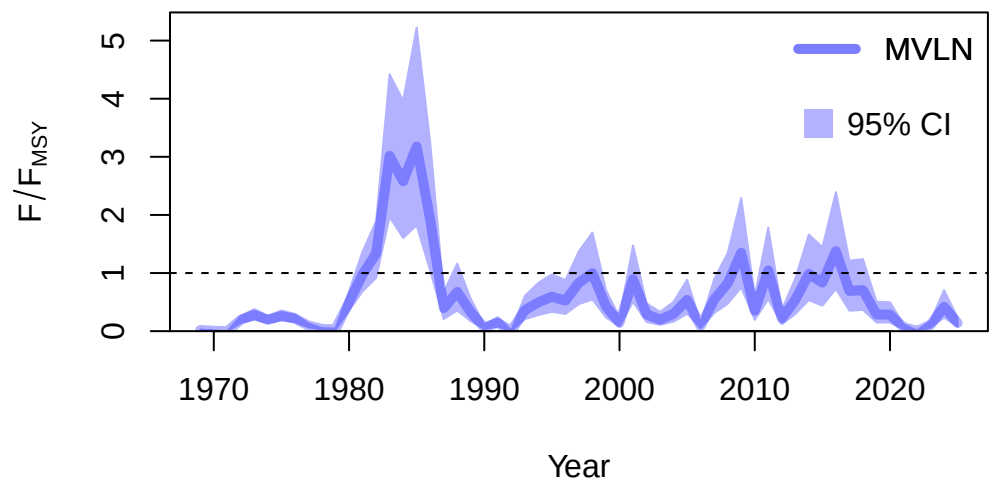
Changes in survey likelihood by fleet

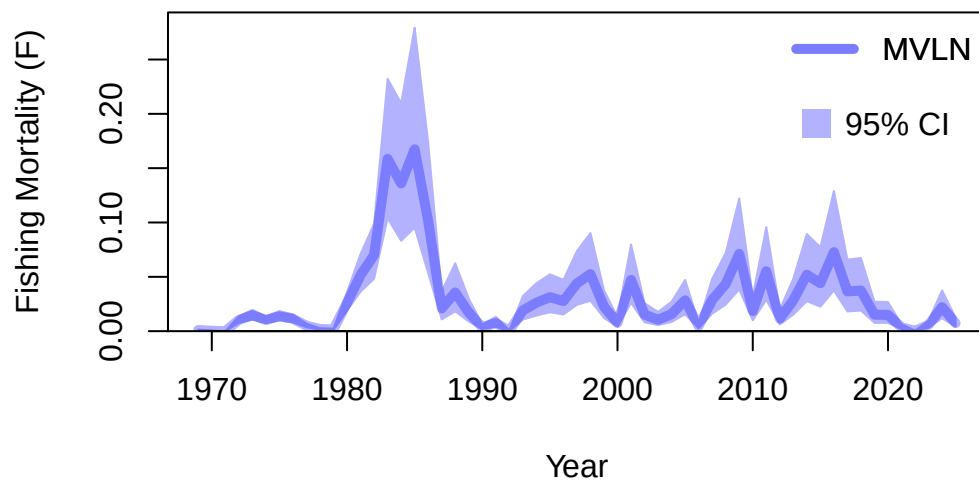


Management Quantities

starter.sso with Bratio: SSB/SSB_{MSY} and F: _abs_F

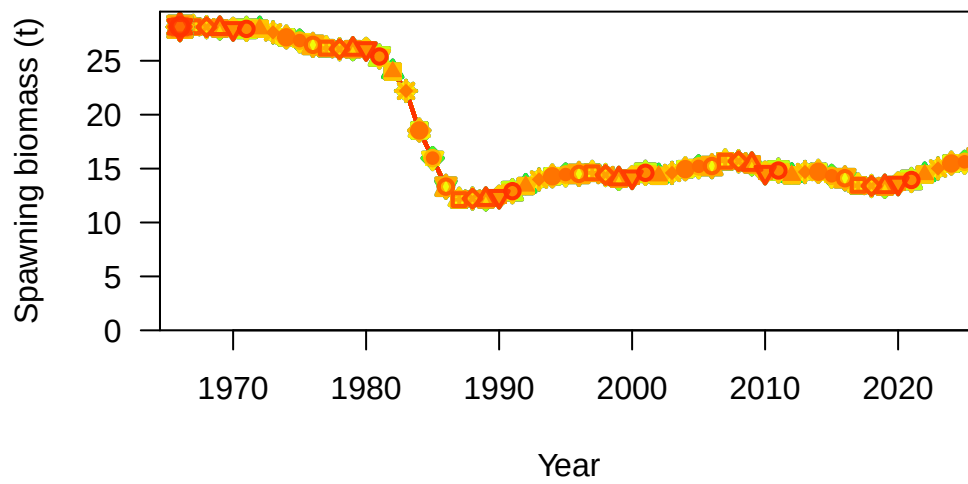
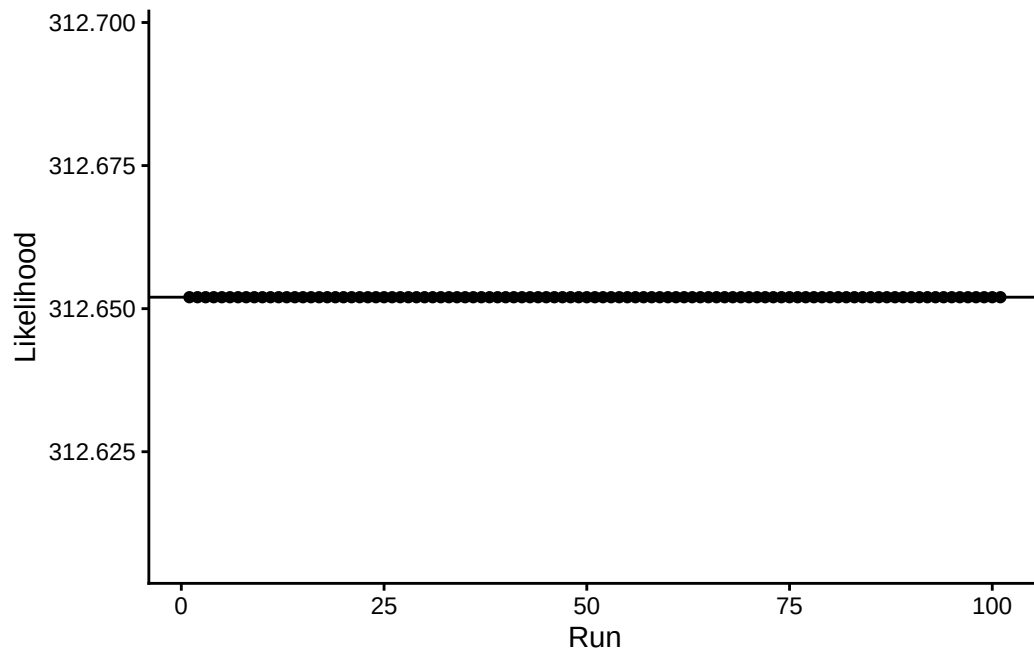


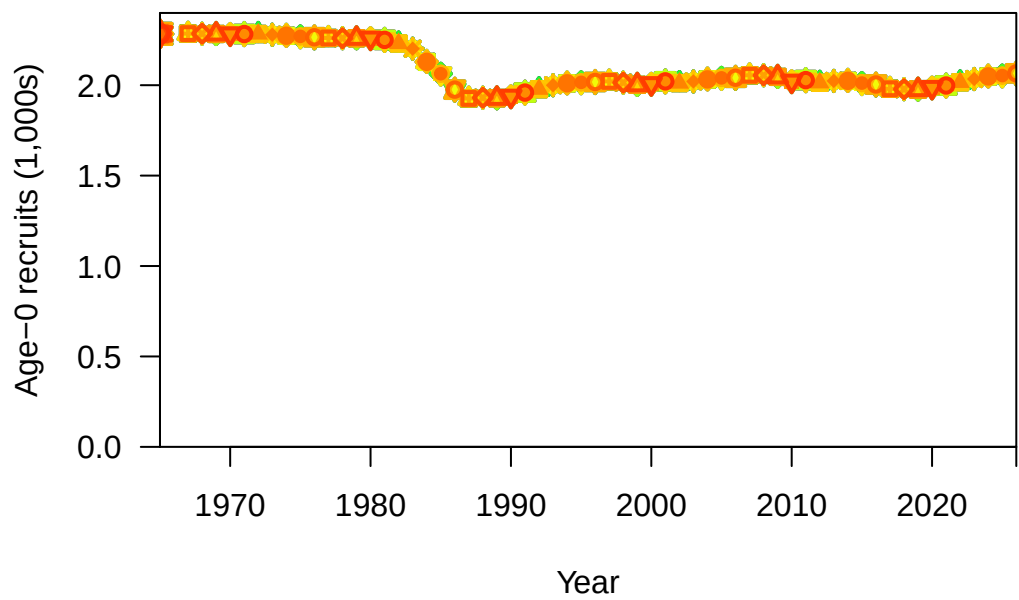
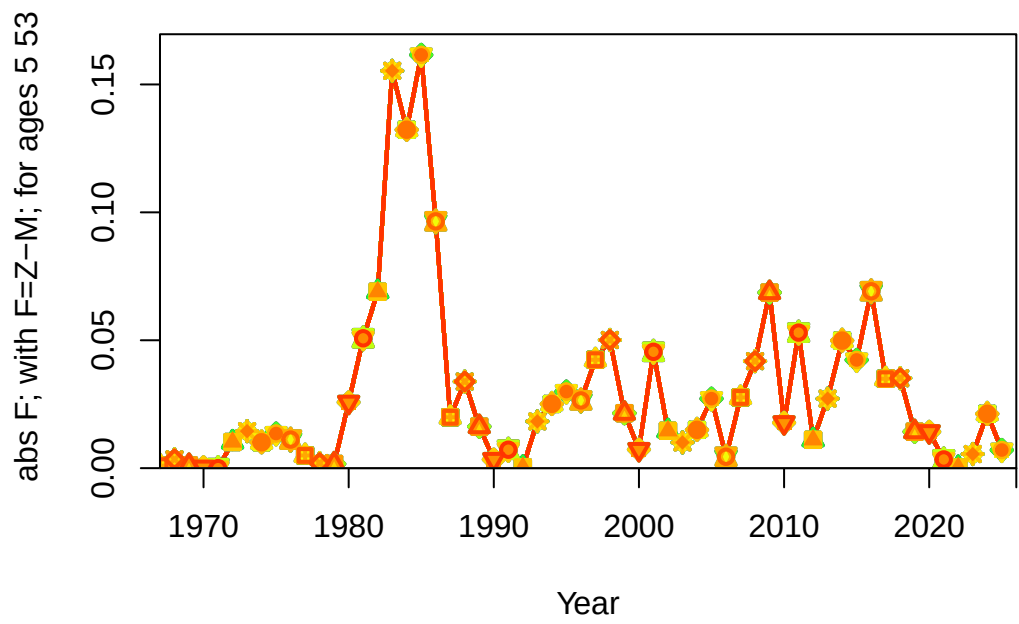




null device
1

Jitter





Selectivity and Maturity

